Extensions of Conway's Game of Life

Erya Du, Faye Jackson, Dianhui Ke, Trey Smith, Benjamin Riley, Cartsten Sprunger, Katie Storey, Jörn Zimmerling

Laboratory of Geometry at Michigan

LOG(M)

Introduction

Basic Settings Conway's Game of Life is a cellular automaton invented by the British mathematician John Horton Conway in 1970 [3]. The universe of the Game of Life is an infinite, two-dimensional orthogonal grid of square cells, each of which is either alive or dead. Every cell interacts with its eight neighbours, which are the cells that are horizontally, vertically, or diagonally adjacent.

Rules

- A living cell with two or three living neighbors survives to the next time step
- A cell with no or one living neighbor dies
- A dead cell with exactly three neighbors will be alive in the next time step

1. Cancer Cell Extension

A Third State:

Probabilistic extension can be further studied for many future works.[1] Here we introduce another state into Conway's Game of Life, which is cancer cell. With the third state, the rules will be slightly different than the original rules.

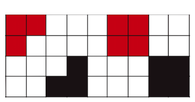
New Rules

There are three states in total: alive, dead, and cancer.

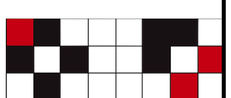
- An alive cell will be non-dead to the next time step, when there are two or three non-dead neighbors. If there are more alive cells than cancer cells around, it will be alive. Otherwise, it will be cancer.
- A dead cell will become alive when there are three non-dead neighbors. If there are more alive cells than cancer cells around, it will be alive. Otherwise, it will be cancer.
- A cancer cell will always be cancer to the next time step when there are between one and three non-dead neighbors. A cancer cell can never be alive unless it dies and becomes alive again.

Example of Stable patterns with Cancer Cells

Blinker(Oscillator)



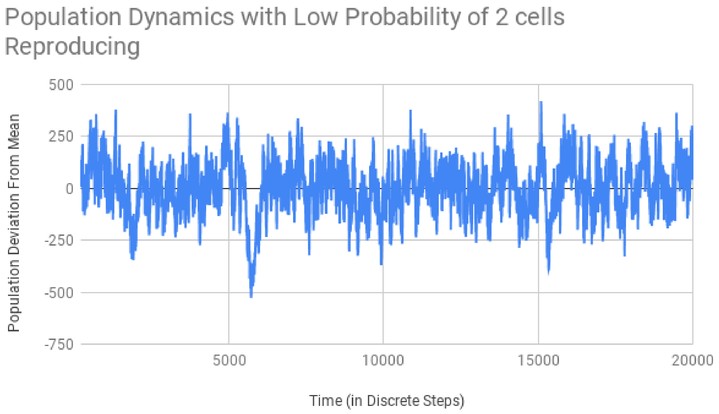
Beacon(Oscillator)

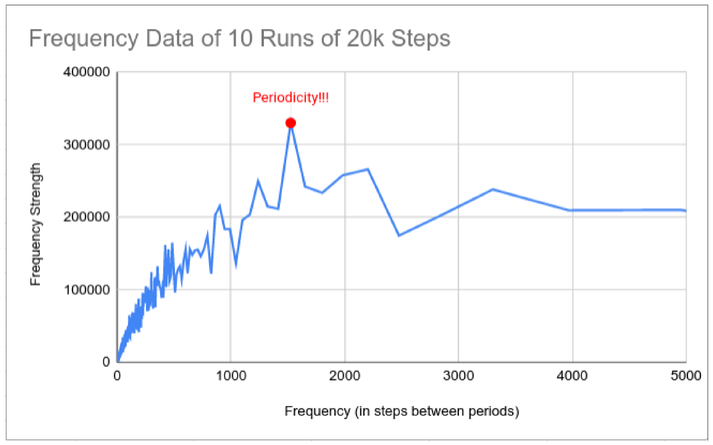


Boat(Still life)

2. Probabilistic Extension

- Heuristic: Most Games of Life with random initial conditions become still lives and oscillators [2]
- Problem: This doesn't fit biology, which we might try to model with Game of Life
- Solution: introduce probabilistic rules to the game of life and look for periodicity in live cell counts over time
- Model: Assign probabilities 0-100% of death, reproduction, or "staying the same" for each number of neighbors. In particular, take probabilities close to the original Game of Life
- Analysis: Use the Discrete Fourier Transform to analyze periodic behavior. Hopefully find periodicity!

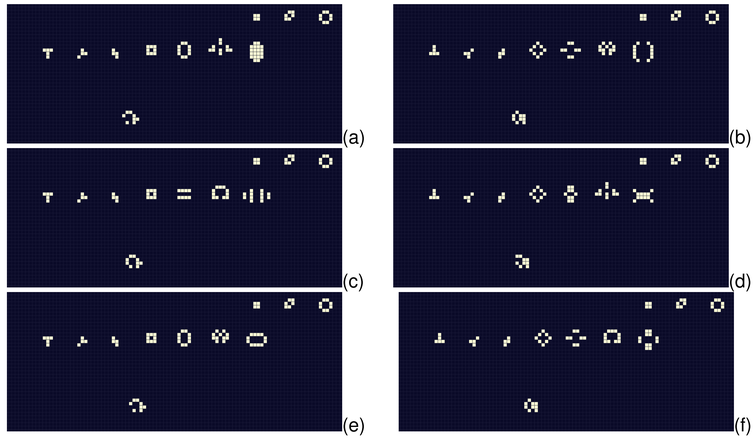




3. Weighted Layers Extension

Example 1 (New Rules with more layers)**.** *Here are the new rules taking into account of more layers outside each single cell. Here is a gallery of the new lives!*

From left to right, Stable Lives:atom,egg,shell
Oscillators: T,Y,Z,Star,Double,Prime,Knot
Gliders: Turtle,Camel



Future Directions

Each of these extensions could be further explored by looking for stable patterns such as still lives, oscillators, and spaceships. In particular, the extension with more layers has many new kinds of patterns that are not in Conway's original game. The cancer cell game could be further refined in order to model real cancer. Finally, the Fourier Analysis method could be used on other probabilistic extensions in order to see if they display periodicity [4].

References

[1] Gabriel Aguilera-Venegas, José Luis Galán-García, Rocío Egea-Guerrero, María Á. Galán-García, Pedro Rodríguez-Cielos, Yolanda Padilla-Domínguez, and María Galán-Luque. A probabilistic extension to Conway's Game of Life. *Adv. Comput. Math.*, 45(4):2111–2121, 2019.

[2] Matthew Cook. Still life theory. 12 2000.

[3] Martin Gardner. Mathematical games the fantastic combinations of john conwayâs new solitaire game âlifeâ. *Scientific American*, 223:120â123, Oct 1970.

[4] Pierre-Yves Louis and Francesca R. Nardi. *Probabilistic Cellular Automata*, volume 27 of *Emergence, Complexity and Computation ECC*. Springer, 2018.